

- 5 (a) Define *magnetic flux density*.

.....

 [1]

- (b) Electrons enter a rectangular block of metal at right-angles to face PQFE, as shown in Fig. 5.1.

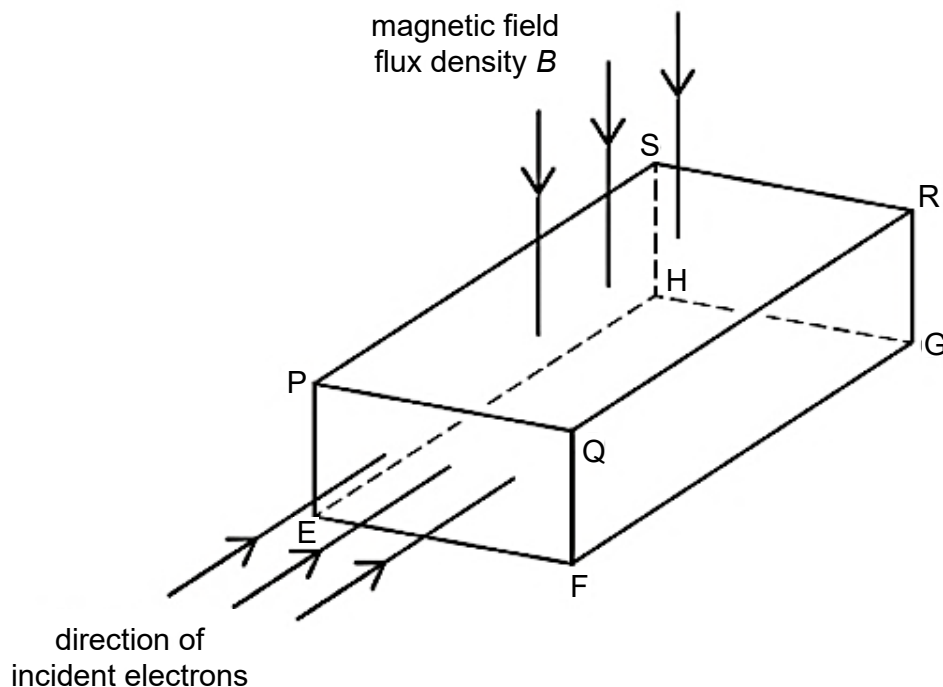


Fig. 5.1

A uniform magnetic field of flux density B is directed at right-angles to face PQRS.

The electrons each have a charge $-q$ and drift speed v in the metal.

Explain why an electron experiences a force due to the magnetic field. State the direction of this force when the electron first enters the block of metal.

.....

 [2]

- (c) After some time, a potential difference is established across two opposite faces of the metal in (b) causing the electrons to travel in a straight line through the metal.

- (i) Explain why a potential difference is established across two opposite faces.

.....

 [2]

- (ii) State the two faces between which the potential difference is established.

face and face [1]

- (d) The potential difference V_H established across the metal in (c) is given by the expression

$$V_H = \frac{BI}{ntq}$$

where I is the current in the metal, B is the magnetic flux density, n is the number of electrons per unit volume, q is the electronic charge and t is the thickness of the metal

- (i) Aluminium has a density of 2.7 g cm^{-3} .
 The mass of one atom of Aluminium is $4.5 \times 10^{-26} \text{ kg}$.
 Assume there is one free electron available to carry charge per atom of aluminium.

Show that the number of charge carriers per unit volume in aluminium is $6.0 \times 10^{28} \text{ m}^{-3}$.

[2]

- (ii) An aluminium foil has a thickness t of 0.090 mm. The current in the foil is 4.6 A.

A uniform magnetic field of flux density 0.15 T acts at right-angles to the foil.

Calculate the potential difference V_H that is generated.

$$V_H = \dots\dots\dots \text{ V [2]}$$

[Total: 10]